Course Code	**OPEN CHANNEL FLOW**	L	T	P	C
23A01604b	**Professional Elective-II**	3	0	0	3

Course Objectives (COs):

The objectives of this course are to make the student to:

1. Explain the principles governing fluid flowin pipelines and networks, including steady and unsteady flow conditions.
2. Apply fundamental concepts of uniform and varied flowin open channels for analyzing hydraulic structures and networks.
3. Analyze the behavior of unsteady flowsin open channels, including wave motion and dam break scenarios.
4. Evaluate sediment transport mechanisms and their impact on hydraulic structures, reservoirs, and river morphology.
5. Design and assess hydraulic models, flow measurement devices, and physical models for hydraulic applications.

Course Outcomes (COs):

After successful completion of this course, students will be able to:

1. Describe the fundamental principles of fluid flowin pipelines and networks under steady and unsteady conditions.
2. Solve problems related to uniform and varied flowin open channels using theoretical and computational approaches.
3. Analyze the impact of unsteady flow phenomena such as surges and dam breaksin open channels.
4. Evaluate sediment transport processes and their influence on river morphology and hydraulic structures.
5. Develop and validate hydraulic models for flow measurement and physical modeling applicationsin fluid mechanics.

CO – PO Articulation Matrix:-

Course Outcomes	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO1**	**PSO2**
CO -1	3	2	-	-	-	-	-	-	-	-	-	-	2	-
CO -2	3	2	2	1	2	-	-	-	-	-	-	-	2	1
CO -3	2	3	2	2	2	-	-	-	-	-	-	-	3	2
CO -4	2	3	3	3	2	-	-	-	-	-	-	-	3	3
CO -5	2	2	3	3	3	2	-	-	-	-	-	-	3	3

UNIT – I

HYDRAULICS of PIPELINES and PIPE NETWORKS

Review of Fluid Mechanics. Reynolds Transport Theorem and Applications. Steady Flow Analysis of Pipe Network Systems. Unsteady Flows - Basic Equations of Water Hammer, Solution By Method of Characteristics. Network Analysis

UNIT – II

STEADY VARIED FLOWS in OPEN CHANNELS

Basic Concepts of Uniform Flow. Specific Energy and Specific force Concepts. Dynamic Equation for Spatially Varied Flows. Flow Profile Computations. IntroductiontoHec-Ras. Spatially Varied Flows and Rapidly Varied Flows – Applications.

UNIT – III

UNSTEADY FLOWS in OPEN CHANNELS

Equations of Motion. Uniformly Progressive Wave. Rapidly Varied Unsteady Flow – Positive and Negative Surges. Dam Break Problem\

UNIT – IV

SEDIMENT TRANSPORT

Sediment Properties – Inception of Sediment Motion – Bed forms. Bed Load Suspended Load – total Sediment Transport. Design of Stable Channels and Regime Channels. Reservoir Sedimentation and Trap Efficiency.

UNIT – V

FLOW MEASUREMENTS and HYDRAULIC MODELING

Sharp-Crested Weirs, Broad-Crested Weirs, Critical Depth Flumes. Recent Advancementin Open Channel Flow Measurements. Physical Modelingin Hydraulics. Dimensional Analysis. Modeling Closed Flows and Free Surface Flows. Distorted Models. Design of Physical Models.

TEXT BOOKS:

1. Flow in Open Channels, Subramanya K., Tata McGraw Hill Pub., N Delhi2015
2. Flow through Open Channels, Rajesh Srivastava, Oxford Univ. Press. N Delhi, 2011
3. OpenChannelHydraulics,Chow,V.T.,McGrawHillInc.NYork,1979

REFRENCE BOOKS:

1. Open Channel Hydraulics, French, R.H., McGraw Hill PubCo.,NYork,1986
2. Open Channel Hydraulics, Terry Sturm, Tata McGraw Hill Pub. N Delhi,2011

Online Learning Resources:

https://nptel.ac.in/courses/105/106/105106114/